

CE 508: Photogeology and Photogrammetry

Course Code	CE 508
Course Name	Photogeology and Photogrammetry
L-T-P-C	2-0-2-3
Prerequisites	None
Intended for	B.Tech. 3rd year/M.S./M.Tech./Ph.D.
Distribution	Elective
Approval	9th Senate (May 2016)

Course Contents

- **Module I**
Basic concepts of Geomorphology, Overview of landscape evolution models, Cycle of erosion, Mountains and relief, river basin, drainage network & types [4 Lectures]
- **Geomorphic landforms, erosional & depositional**
Fluvial, Glacial, Aeolian, Coastal and Karst landforms. [6 Lectures]
- **Module III** Overview of Indian geomorphology. [2 Lectures]
- **Module IV**
Photogrammetry introduction, types of aerial photograph, Geometry, Scale and resolution its applications. [5 Lectures]
- **Module V**
Principles of stereoscopy, lens and mirror stereoscopes, image parallax, relief displacement, vertical exaggeration, distortion. Measurement of relief displacement and estimation of height of an object. [10 Lectures]
- **Airphoto interpretation & mapping**
Identification of various rock types and landforms. Delineation and mapping of various geomorphic features (Fluvial, Glacial, Aeolian, Coastal), rock types (Igneous, Sedimentary and Unconsolidated sediments) and structural features (Fold, Faults, Joints, Lineaments, Synclines & Anticlines). [12 Lectures]
- **Module VII**
Introduction to satellite image processing, geoinformatics, DEM. [3 Lectures]

Text Books

1. A L. Bloom, **Geomorphology: A, systematic Analysis of Late Cenozoic Landforms**, 3rd Edition, Pearson Education, Inc., USA, 2004.
2. V S. Kale, and A. Gupta, **Introduction to Geomorphology**, Orient Longman Ltd., India, 2001.
3. Victor, C. Miller, **Photogeology**, McGraw Hill Book Co., New York, 1961.
4. P R. Wolf and B A. Dewitt, **Elements of Photogrammetry: With Applications in GIS**, McGraw Hill Science, New York, 2014.
5. T.M. Lillesand and R.W. Kiefer, **Remote Sensing and Image Interpretation**, 6th Edition, John Wiley & Sons, New York, 2007.

References

1. J R. Jensen, **Remote Sensing of the Environment an Earth Resource Perspective**, Pearson Education. Delhi, 2003.
2. Julien, P.Y., **River Mechanics**, Cambridge University Press, USA, 2002.
3. G R Davis, SR Reynolds, CF Kluth, **Structural Geology of Rocks and Region**, 3rd Edition, John Wiley, USA, 2012.
4. M P. Billings, **Structural Geology**, 4th Edition, Prentice-Hall, New York, 1987.